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5 / 5 submitted

Q1. Exception

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int[] arr = {10, 20, 30, 40, 50};

        int index = sc.nextInt();

        try{
            System.out.println(arr[index]);
        }catch(ArrayIndexOutOfBoundsException e){
            System.out.println("Exception: Invalid array index.");
        }
    }
}
```

INPUT

6

OUTPUT

Exception: Invalid array index.

Q2. IllegalArgumentException

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int marks = sc.nextInt();

        try{
            if (marks < 0 || marks > 100){
                throw new IllegalArgumentException();
            }
            System.out.println("Valid marks: " + marks);
        }catch(IllegalArgumentException e){
            System.out.println("Exception: Invalid marks. Marks should be between 0 and 100.");
        }
    }
}
```

INPUT

120

OUTPUT

Exception: Invalid marks. Marks should be between 0 and 100.

Q3. user-defined exception named WeakPasswordException.

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.*;
class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        String password = sc.nextLine();

        try{
            if(password.length() < 8){
                throw new WeakPasswordException("Password must contain at least 8 characters.");
            }else{
                System.out.println("Password is strong.");
            }
        }catch (WeakPasswordException e){
            System.out.println("Exception: " + e.getMessage());
        }
    }
}
class WeakPasswordException extends Exception{
    WeakPasswordException(String s){
        super(s);
    }
}
```

INPUT

java123

OUTPUT

Exception: Password must contain at least 8 characters.

Q4. Email Validation

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.*;
class Main{
    public static void main(String[] args){
        java.util.Scanner sc = new java.util.Scanner(System.in);

        String email = sc.nextLine();

        try{
            if(email.indexOf("@") == -1){
                throw new InvalidEmailException("Invalid email address.");
            }

            System.out.println("Valid email address.");
        } catch(InvalidEmailException e){
            System.out.println("Exception: " + e.getMessage());
        }
    }
}
class InvalidEmailException extends Exception{
    InvalidEmailException(String s){
        super(s);
    }
}
```

INPUT

student@gmail.com

OUTPUT

Exception: Invalid email address.

Q5. Banking Transaction Using Exception Propagation

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        sc.next();
        int balance = sc.nextInt();
        sc.next();
        int withdrawAmount =sc.nextInt();
        try{
            withdraw(balance,withdrawAmount);
        }catch(InsufficientBalanceException e){
            System.out.println("Exception propagated to main().");
            System.out.println("InsufficientBalanceException: "+e.getMessage());
        }
    }
    static void withdraw(int balance,int withdrawAmount)throws InsufficientBalanceException{
        validateBalance(balance,withdrawAmount);
    }
    static void validateBalance(int balance,int withdrawAmount)throws InsufficientBalanceException{
        processTransaction(balance,withdrawAmount);
    }
    static void processTransaction(int balance,int withdrawAmount)throws InsufficientBalanceException{
        if(withdrawAmount>balance){
            throw new InsufficientBalanceException("Withdrawal amount exceeds available balance.");
        }
    }
}
class InsufficientBalanceException extends Exception{
    InsufficientBalanceException(String message){
        super(message);
    }
}
```

INPUT

Balance:
50000
Withdraw:
70000

OUTPUT

Exception propagated to main().
InsufficientBalanceException: Withdrawal amount exceeds available balance.